Reading a rooftop against the sky

Nine months of electricity bills, ten years of satellite weather, and what they say about one bifacial solar array in Bhubaneswar.

By Piyush Dash · October 2026 · interactive version and code: github.com/piyushdash94/rooftop-solar-analytics

In February 2026 a 3 kWp array of bifacial panels went up on the roof of our family home in Satya Vihar, Bhubaneswar. Eight months later we had a folder of bills and a few questions that sounded simple. Is the system working as it should? Why did the bills go up when the panels were new? What will it make next year?

The data. Nine monthly TPCODL bills, 8 February 2026 to 1 November 2026, each listing cumulative readings from the solar generation meter and the import–export net meter; a month's units are the difference between two consecutive readings.

We had no inverter app logging output minute by minute, only those monthly numbers. This piece is about how far they can go once you join them to the weather, and about the hunches we had to test, drop or keep along the way.

3,181 kWh
generated over 267 days, 8 February 2026 to 1 November 2026.

91%
of the month-to-month variation is explained by sunlight alone.

−1.1%
miss against the October forecast we published before the bill arrived. Its typical miss across all months is about 5%.

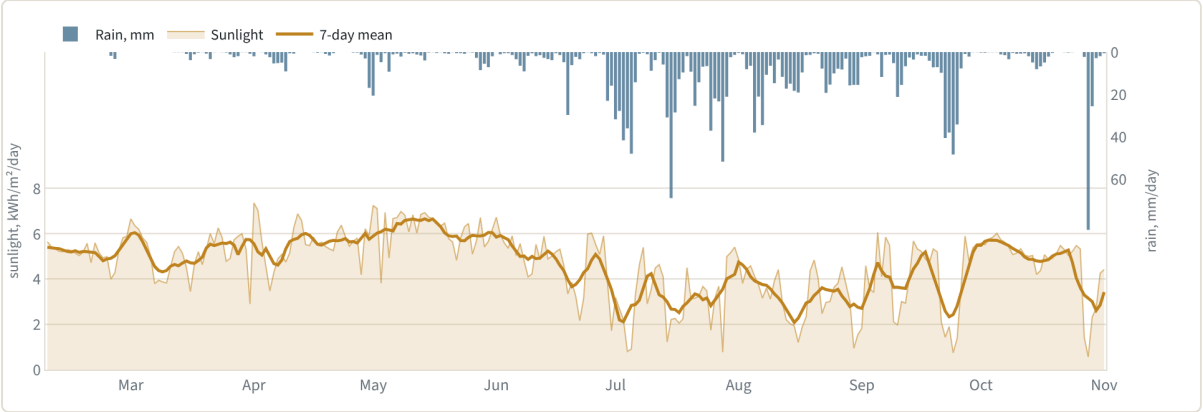
4,518 kWh
expected in a typical weather year. The last ten years of weather range from 4,254 to 4,643.

₹19,926
saved on bills so far.

+506 kWh
a year, modelled, if the roof under the panels is painted white.

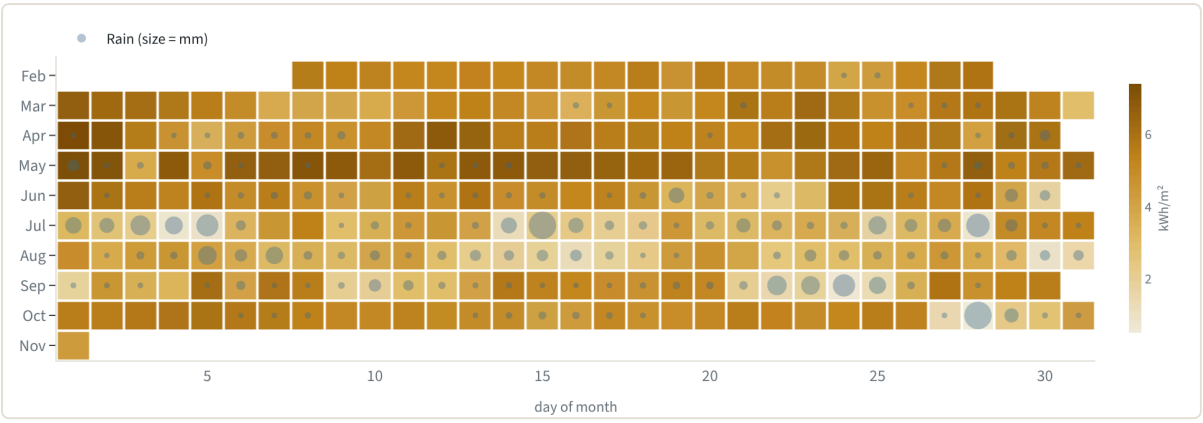
Hunch one: the sun does nearly all the work

The first hunch was the obvious one: more sun, more power. To test it we needed the sunlight that fell on this roof on each billed day. Bills do not carry that, so we took it from NASA's POWER service [1]. For any latitude and longitude, POWER returns a daily estimate of the solar energy reaching a horizontal surface at the ground, in kilowatt-hours per square metre per day (the parameter called ALLSKY_SFC_SW_DWN). The estimate comes from satellite measurements of clouds and the atmosphere, processed by NASA's CERES programme on a grid of 1° by 1°, about 110 km on a side. Air temperature and rainfall come from NASA's MERRA-2 reanalysis on a finer grid. We requested daily values for 20.312° N, 85.868° E through the POWER web API; it is free, needs no key, and runs a few days behind real time.



Daily sunlight (left axis, with its 7-day mean) and rainfall (right axis, hanging from the top) for every billed day. NASA POWER, at the roof's coordinates.

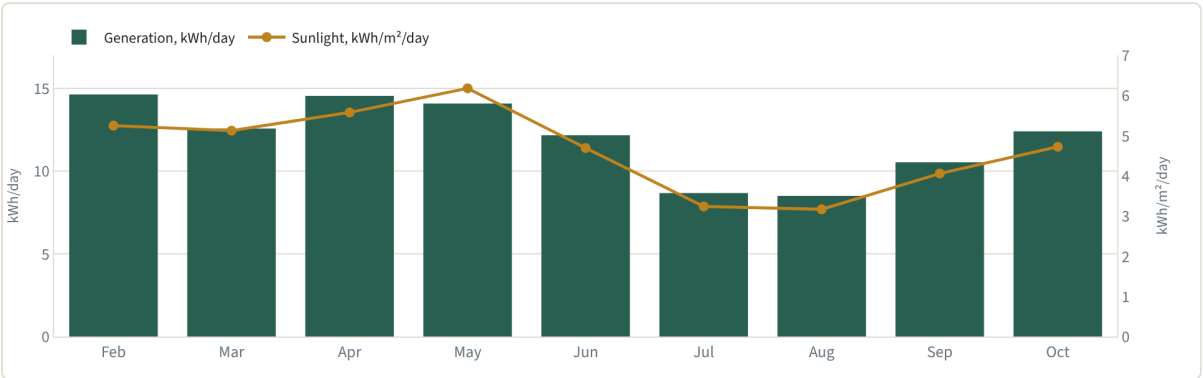
The shape of the year is plain. Through spring the roof received 5 to 6 kWh of sunlight per square metre each day. Rain builds from the second week of June, and by July the daily average was close to 3.



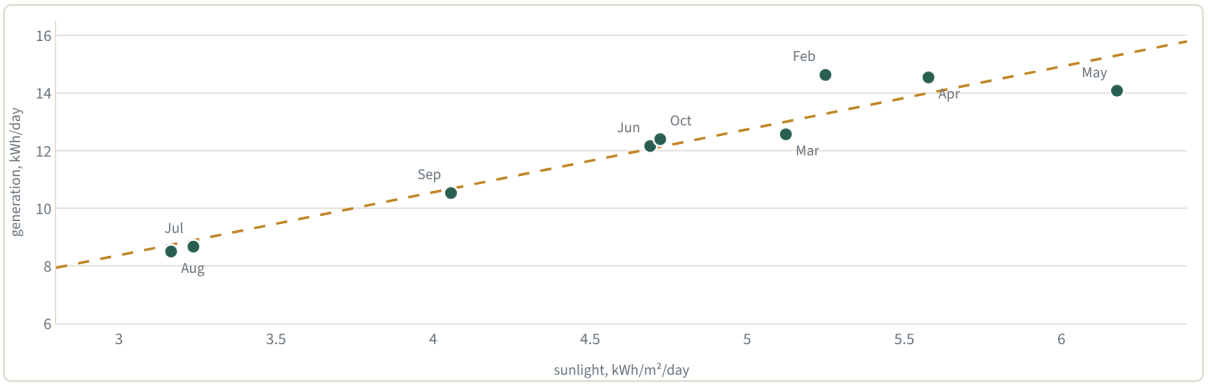
Each cell is one day: deeper colour means more sunlight; blue dots mark rain, sized by millimetres. The monsoon arrives as a run of pale cells from mid-June.

Each billing month then becomes two numbers. **Generation per day** is the month's metered generation divided by the number of days in the billing period. **Sunlight per day** is the average of NASA's daily values over exactly the same dates. For the first bill, that is 321.9 kWh over 22 days, or 14.6 kWh a day, against an average of 5.25 kWh/m² of sunlight a day. Dividing by days matters because billing periods run from 22 to 31 days.

Plotted month by month, the two move together almost in a straight line.



Bars: each billing month's generation ÷ its number of days. Line: NASA's daily sunlight averaged over the same days.



Each dot is one billing month. The dashed line is the least-squares fit: generation per day = a × sunlight + b.

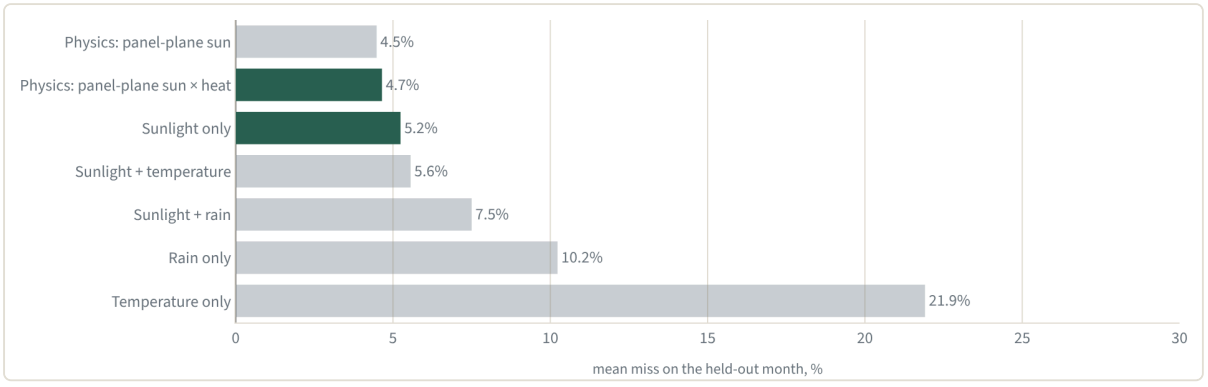
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generation per day = 2.18 × sunlight + 1.82
(generation in kWh/day; sunlight in kWh per m² per day)
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Sunlight accounts for 91% of the variation between months ($R^2 = 0.91$). Everything else, including dust, heat, shading and power cuts, has to fit in the remaining 9%.

Choosing what to leave out

A line through nine points is easy to draw and easy to fool yourself with. With so few months, every extra variable is another way to fit noise. So instead of asking which features sounded reasonable, we asked which ones helped predict a month the model had not seen.

The test is called leave-one-out. Fit the model on all months but one, predict the one left out, and repeat for every month. The average miss is the number to beat.



How far each candidate model missed the one month it was not allowed to see, averaged over all nine months. Green: the two models kept.

Sunlight alone missed by 5.2% on average. Adding temperature made it worse, at 5.6%, and so did adding rain, at 7.5%. Rain on its own (10.2%) and temperature on its own (21.9%) were far worse. Rain matters, but its effect is already inside the sunlight number, because rainy days are dull days. Adding it a second time only gives the fit another way to chase noise.

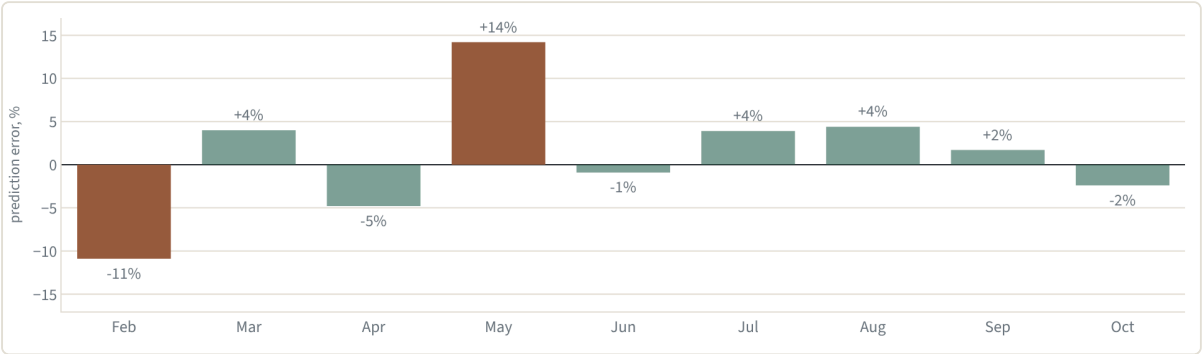
We kept two models. The simple sunlight line is the quick one: one input, two fitted numbers. The physics model described below misses by 4.7% on average, slightly better than the line. When we first ran this comparison the order was reversed, until NASA revised its July and August data (see below). With nine points, a gap that small is not a ranking. We keep the physics model for two things the line cannot do: judge the panels' health month by month, and forecast winter months the line has never seen.

Every analysis is a list of decisions about what to ignore. Here is ours.

Feature	Decision	Why
Sunlight on the ground	Kept	Explains 91% on its own.
Panel angle	Simplified	Fixed at the installed 12°, facing south. Used to convert horizontal sunlight into sunlight on the panels.
Direct and diffuse light	Simplified	Split with a standard daily formula, because there is no hourly data.
Panel temperature	Kept, physics model only	The datasheet gives a 0.30% loss per °C, and panels run hotter in strong sun. Monthly data is too coarse to see it clearly, but physics says it is there.
Rain	Left out	Its effect is already in sunlight. Adding it worsened predictions.
Cloud cover	Left out	Carries the same information as sunlight.
Humidity and wind	Left out	No measurable signal at monthly resolution.
Dust and soiling	Not modelled	Appears in the residuals instead.
Power cuts	Not modelled	No logs. The family estimates about 5 hours a month in daylight.
Ground reflectivity	Fixed at 0.12	Dark concrete. Changed only in the white-paint scenario.
Panel ageing	Ignored for 8 months	About 0.4% a year. Included in the lifetime outlook.
Who is at home	Demand only	The house stood empty for about a week in May and had guests for about a week in October. That changes what the house used, never what the panels made.

Did the line see the future?

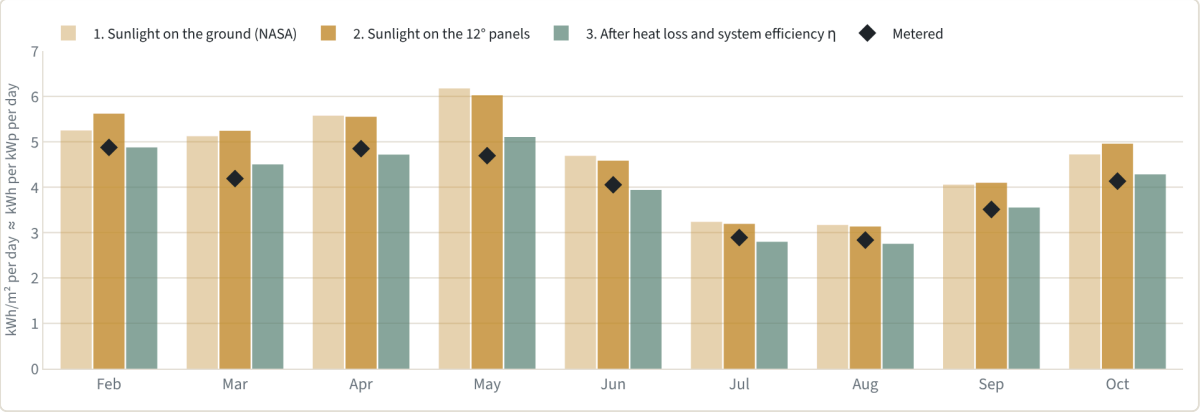
The cleanest test of a model is a month that did not exist when it was built. We published a forecast before October's bill arrived: 389 kWh for a typical October. The meter recorded 384, a miss of −1.1%. The sunlight line, fitted on February to September and given October's measured sunlight, predicted 375 kWh, a miss of +2.5%, well inside its usual scatter of about ±23 kWh for a month that long. September, tested the same way a month earlier, missed by under 4%.



Fit on all months but one, predict the one left out, repeat for every month. September and October were also genuine out-of-sample tests.

One good result can be luck, so the leave-one-out chart is the fairer picture. A typical month lands within about 5%. The two worst misses point in opposite directions: February was under-predicted by 11% and May over-predicted by 14%.

Why February was under-predicted. The line only knows the sunlight that reaches flat ground. The panels are not flat, and they are not at air temperature. In February the sun stays low in the southern sky, so panels tilted 12° towards it catch it more squarely: they received 7.1% more sunlight than flat ground. In May the sun is almost overhead at noon, and the same tilt costs a little: 2.5% less. February was also cooler (a mean of 24 °C against 32 °C in May), so the cells lost 5.0% of their output to heat against 7.2% in May. Put together, each unit of ground-level sunlight was worth about 12% more electricity in February than in May. That is the size of both misses. The chart below follows each month from sunlight to kilowatt-hours and shows where the two months part company.



From sunlight to kilowatt-hours, month by month. Because 1 kWh/m² on the panels gives about 1 kWh per kWp at test conditions, all bars share one scale. Hover for the geometry gain and heat loss.

The physics model, input by input

To judge the panels themselves we needed an estimate of what the weather allowed each month, built from physics rather than fitted to the bills. It runs one day at a time and adds the days up for each billing period.

```
expected energy = sunlight on the panels × heat derate × 3 kWp × η
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Step 1: sunlight on the panels. NASA gives sunlight on flat ground. For each day we first compute how much sunlight would arrive at the top of the atmosphere at this latitude on that date; the ratio of the two, the clearness index, says how clear the day was. A standard formula from Erbs, Klein and Duffie [2] turns the clearness index into the share of light that was diffuse, scattered across the sky, rather than direct from the sun. Direct light is then projected onto panels tilted 12° to the south, using the sun's position through the day [3][4]. Diffuse light is counted from the part of the sky the tilted panels can see [5], and a small amount of light is added for reflection off the roof.

Step 2: heat. These modules lose 0.30% of their output for every degree their cells run above 25 °C [6]. Cell temperature is estimated with the datasheet's nominal operating cell temperature (NOCT) of 43 °C, the temperature the cells reach under 800 W/m² of sunlight with the air at 20 °C [6][4]. Cells warm above the air in proportion to the sunlight on them, so the panels run hot in clear spring weather and stay near air temperature under monsoon cloud.

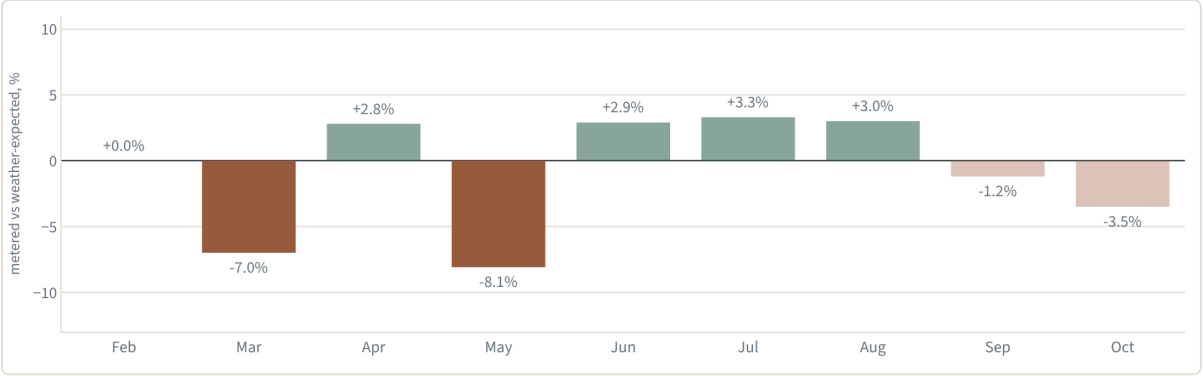
Step 3: one fitted number. One kilowatt-hour of sunlight per square metre on the panels would give about one kilowatt-hour per kilowatt-peak at test conditions. Real systems lose some of that in the inverter, the wiring, light reflected off the glass, and dust. We let the data set that single factor, η, as the median of the nine months'

metered generation divided by the energy the first two steps predicted. It comes out at 0.913.

Input	Value	Where it comes from
Daily sunlight on the ground	kWh/m² per day	NASA POWER [1]
Daily mean air temperature	°C	NASA POWER, MERRA-2 [1]
Site location	20.312° N, 85.868° E	The roof's coordinates
Panel tilt and direction	12°, facing south	The installed angle, taken as fixed
Ground reflectivity	0.12	Assumed for dark, often damp concrete
Power loss with temperature	−0.30% per °C	Module datasheet [6]
NOCT	43 °C	Module datasheet [6]
Array rating	3 kWp	The installation
System efficiency, η	0.913	Fitted: median of metered ÷ predicted over the nine months

Because η is fitted on the same months the model judges, the model can only compare months with one another. It cannot say how good the system is in absolute terms.

Two months that fell short



Metered generation against what the physics model expected from the weather. Hover for the kWh.

March (−7.0%) and May (−8.1%) are the only real shortfalls.

March is the story of a hunch that was wrong. Early on we flagged it as suspicious, reasoning that March in Odisha is usually drier and sunnier than April. The satellite record said otherwise: March 2026 received less sun than April. Most of March's low output was weather, and only the residual in the chart survived once the weather was accounted for.

May had the most sun of any month, the highest mean air temperature and the worst conversion, even after the model's heat correction. Heat beyond what the model captures, and pre-monsoon dust, are the candidates. April and June, which followed 75 and 162 mm of rain, both came in above expectation (+2.8% and +2.9%), which fits dust being washed off. But May itself had 59 mm of rain and still fell short, so monthly bills cannot separate the two. The experiment further down tries.

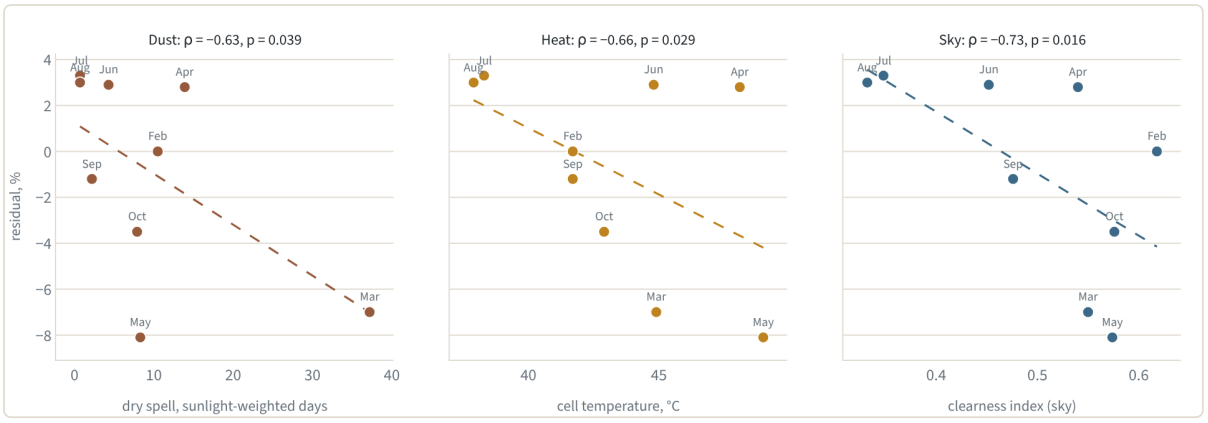
July and August came in +3.3% and +3.0% above expectation, within the model's usual scatter. Our first version showed them 7 to 10% high, and we blamed the daily formula that splits direct from diffuse light under heavy cloud. That explanation was wrong. NASA later revised its July sunlight up by about 5% and August by about 9%, and the excess disappeared. The early satellite estimates had under-read the monsoon.

An experiment: dust, heat or sky?

The shortfalls left a question. Was it dust on the glass, heat the model under-counts, or something about clear skies? Each explanation leaves its own fingerprint in the data, so each can be tested.

- **Dust.** Dust builds up between rains and washes off in a downpour. For every day we counted the days since the last rain of at least 5 mm, starting from zero on the day the new panels went up, and averaged that over each billing month, giving bright days more weight. Prediction: the longer the dry spell, the lower the residual.
- **Heat.** If the model's heat correction is too gentle, the hottest months will still fall short. Proxy: the modelled cell temperature. Prediction: hotter cells, lower residual.
- **Sky.** If the model treats clear and cloudy days unevenly, the residual will track how clear the sky was. Proxy: the clearness index, measured sunlight ÷ clear-sky sunlight.

We tested each on the 9 months with a rank correlation and an exact permutation test. The test asks how often a random shuffle of the months would give a correlation at least this strong. With nine points, it makes no assumptions we cannot check.



Each dot is one of the nine discovery months: its residual against a dust proxy (days since rain, weighted by sunlight), cell temperature, and sky clearness. ρ is the rank correlation; p is an exact one-sided permutation test.

All three point the same way: dust ρ = −0.63 (p = 0.039), heat −0.66 (p = 0.029), sky −0.73 (p = 0.016). Running three tests raises the odds of a lucky hit, so we corrected for it (Holm). Only the sky clears 0.05, and only just, at 0.047. Dust and heat sit at 0.059.

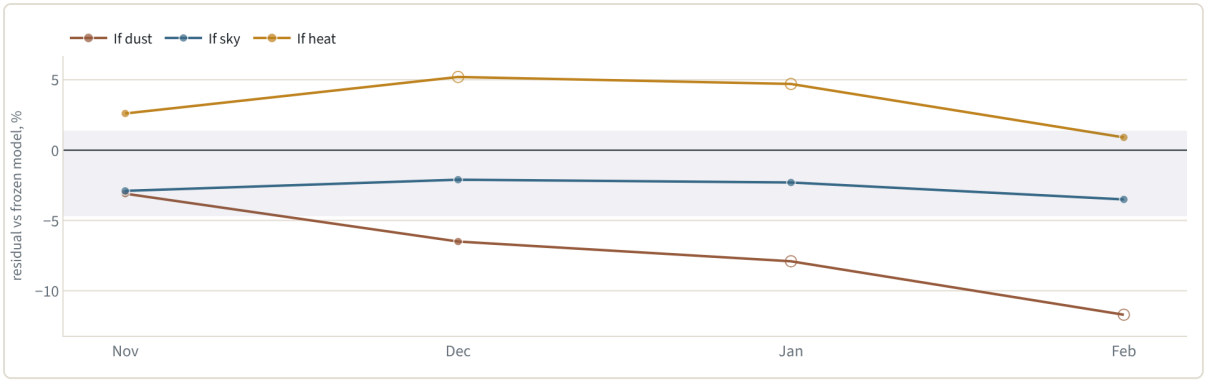
The trouble is that the three travel together. In Bhubaneswar the dry months are also the clear months and the hot months. The dust proxy correlates +0.71 with clearness and +0.75 with cell temperature. So we asked which one survives when another is held fixed. Holding clearness fixed, the dust correlation shrinks to −0.22. Holding dust fixed, clearness keeps −0.52. Heat also keeps −0.52 with clearness held fixed. Dust is the weakest of the three. Heat and sky cannot be told apart with these months.

April and May show the limit. Their cells ran equally hot and their skies were equally clear, yet April came in at +2.8% and May at −8.1%. None of the three proxies separates them, so part of May's shortfall remains unexplained.

A forecast to settle it

Winter pulls the three apart. From November to February the sky stays clear, the cells run cool and it rarely rains, so the dry spell grows from about 20 days to about 58. Feeding the typical winter weather of 2016–2025 into each fit gives three different forecasts.

- If **dust** drives the residual, it should sink month by month, from −3.1% in November to about −11.7% in February. December–January average: −7.2%.
- If **heat** drives it, cool winter cells should push it up. December–January average: +5.0%.
- If **sky** drives it, it should stay flat. December–January average: −2.2%.

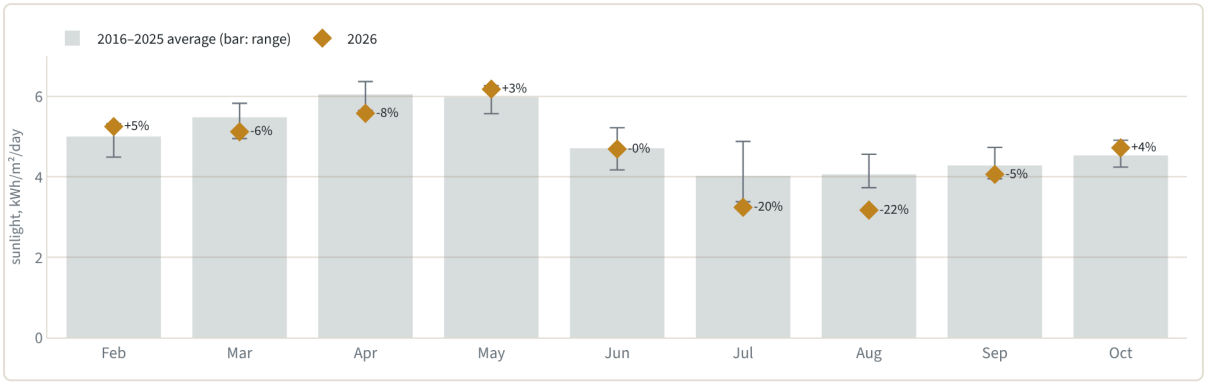


What each explanation predicts for the winter bills, written down on 8 November 2026. Open circles fall outside the range the fit was made on. Shaded: the band where the verdict is 'sky'. Metered diamonds appear as bills arrive.

We wrote this down on 8 November 2026, before any winter bill. The model's efficiency is frozen at 0.913, so new months cannot shift the yardstick. The verdict uses the average of the December and January residuals. Below −4.7% favours dust, above +1.4% favours heat, and anything in between favours sky. A fall of five points or more from November to February also counts for dust. Two months carry noise of about ±3.1 points, so there is roughly a 21% chance of mistaking sky for dust and a 13% chance of mistaking it for heat. The dust and heat fits also reach beyond the range they were built on (open circles), which is why the rule rests on the order of the three lines rather than their exact values. The test assumes nobody washes the panels this winter.

2026 was not a normal year

Our first projection for the next twelve months simply reused 2026's weather. That turned out to be a mistake worth explaining.



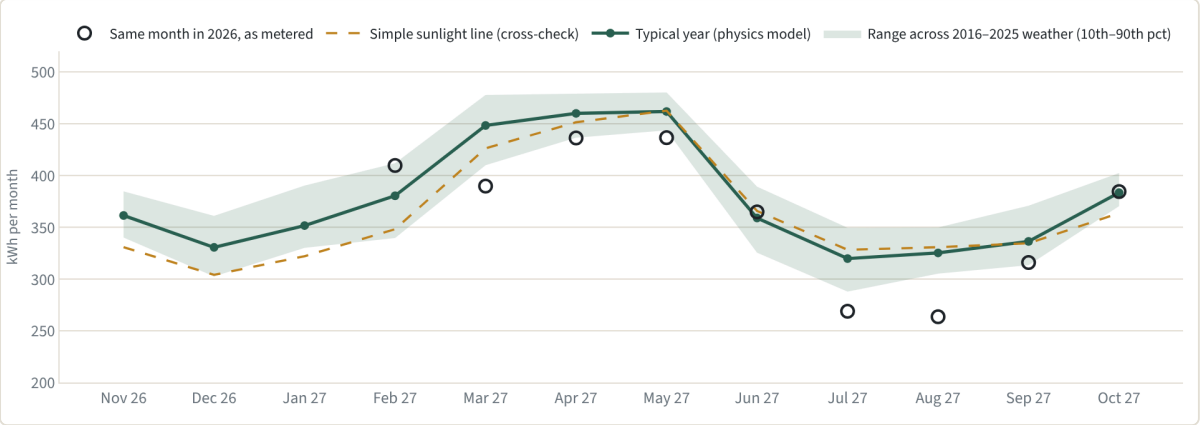
Sunlight in each 2026 billing month (diamonds) against the same calendar dates in 2016–2025 (bars, with the full range).

Comparing each billing month with the same calendar dates over the previous ten years shows how unusual 2026 was. July and August received −20% and −22% less sunlight than normal, darker than any of the previous ten years. July's rainfall was 563 mm against a ten-year average of 318 mm. Across the nine billed months, sunlight ran −5% below normal.

NASA revises its most recent months as processing completes. June changed by about 2% between downloads, and July and August by 5 to 9%, so October may still shift. The rainfall, which comes from a separate dataset, tells the same story.

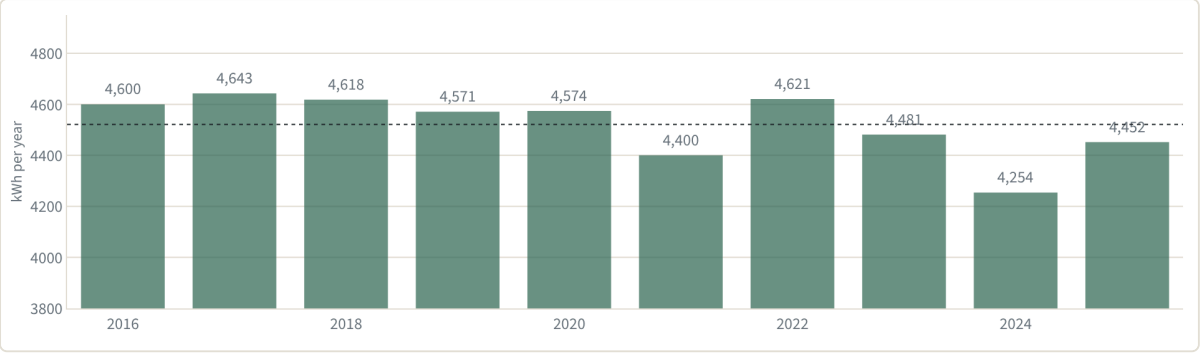
What a normal year should look like

So instead of projecting from one unusual year, we ran the physics model over every day of the last ten years of weather at the site (2016 to 2025), with η fitted on 2026.



Expected generation for the next twelve months: the physics model run on each of the last ten years of weather. Open circles show what the same calendar month produced in 2026.

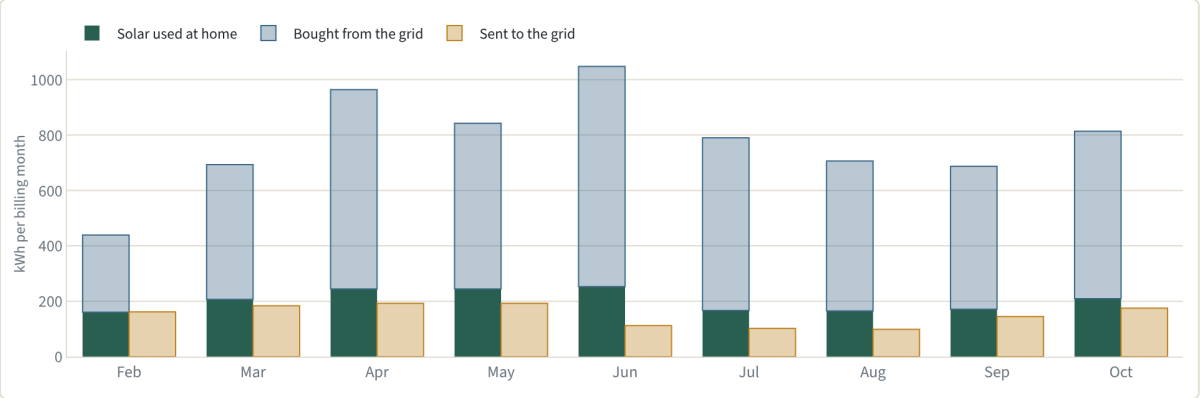
A typical year comes to about 4,518 kWh for Nov 26 to Oct 27. The simple sunlight line gives 4,368 kWh for the same months, about 3% lower, and most of the gap falls in the low-sun months from October to March. That is the February effect again. In winter the sun is low, and panels tilted towards it collect more than flat-ground sunlight suggests. The line has only seen February to October and has never seen a deep-winter sun. The physics model knows the geometry.



The same array, run through each of the last ten years of weather. Dotted line: the ten-year average.

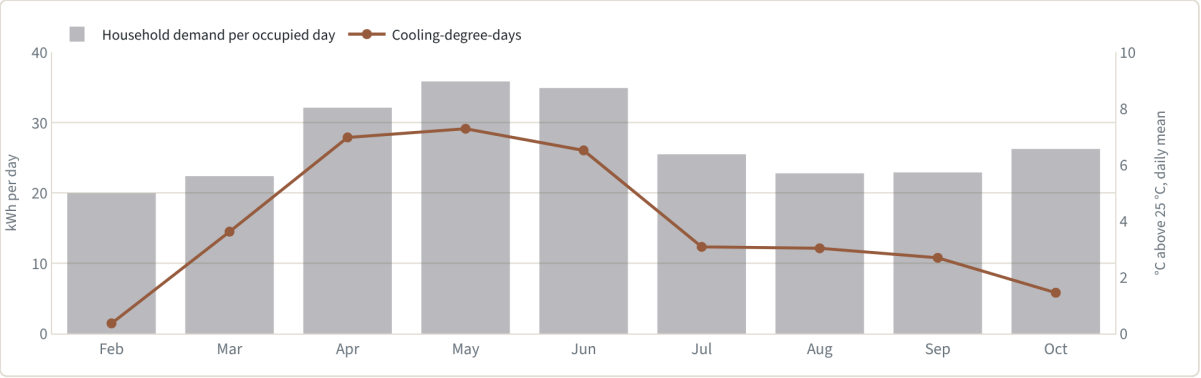
Year to year, the weather moves the total by only 2.7%. The worst of the last ten years, 2024, would have given 4,254 kWh; the best, 2017, 4,643 kWh. A bad year costs a few percent, not a fifth.

Where the units went, and what they were worth



Each month's household demand (stacked: solar used at home plus electricity bought) beside what was sent back to the grid.

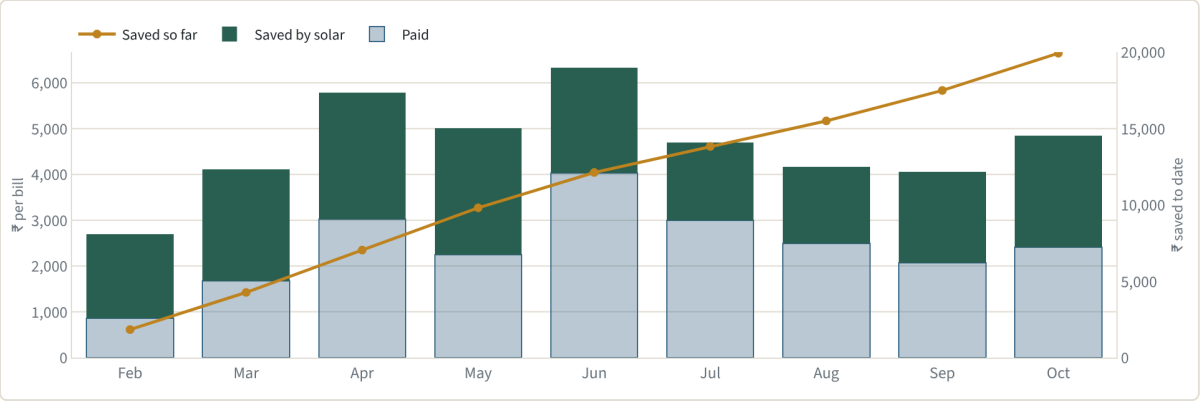
Solar used at home was steady, at 160 to 253 kWh a month. Demand was not. It rose about 75% per day from February to June (20 to 35 kWh a day) as the air conditioning came on, then eased with the rains. The bills went up because the house used more, not because the panels made less. A 3 kWp system against a 1,000 kWh month is undersized for summer rather than underperforming.



Household demand per occupied day (bars) and cooling-degree-days (line). May's days are counted net of the week the house stood empty.

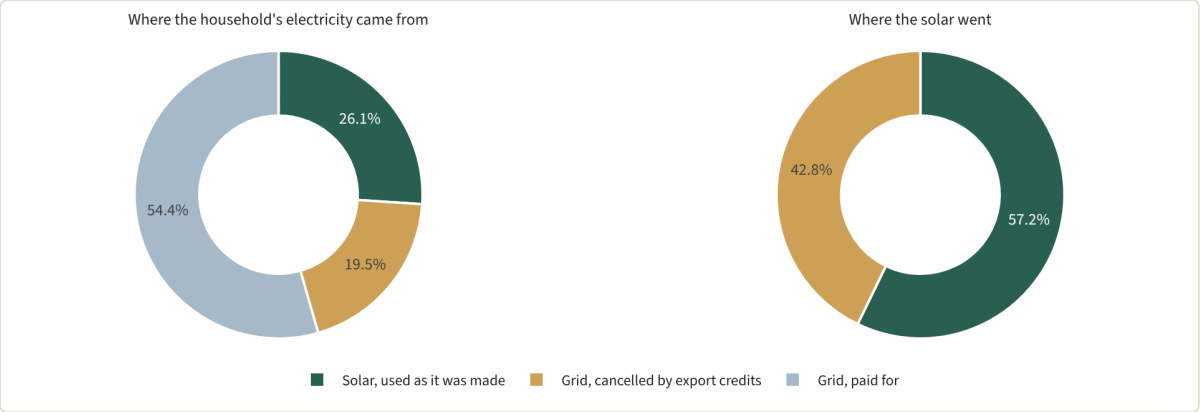
Demand follows the heat. Measured by cooling-degree-days, how far the daily mean temperature sat above 25 °C, the correlation is +0.90. Demand here is counted per occupied day, because the house stood empty for about a week in May, and the family recalls importing roughly 200 fewer units than usual. An absence of 7.5 days reproduces that figure, so May's demand is spread over its occupied days only.

October broke the pattern the other way. It was cooler than September, yet the house used 15% more per day (22.9 to 26.2 kWh). We had guests for about a week. Fitted on the other months, heat alone would predict 20.4 kWh a day for October, which leaves about 180 kWh unexplained. If the guests account for all of it, that is roughly 26 kWh a day for their week, more than the house normally uses in a day, so the bills cannot say how much belongs to them. With October set aside, the correlation with heat is +0.95.



What each bill cost and what solar saved, slab by slab under net metering, with the running total.

Under net metering, every exported unit is credited against an imported one, so the bill follows imports minus exports. Comparing each bill with what it would have been without solar, slab by slab at the domestic tariff [7], solar has saved ₹19,926 so far.



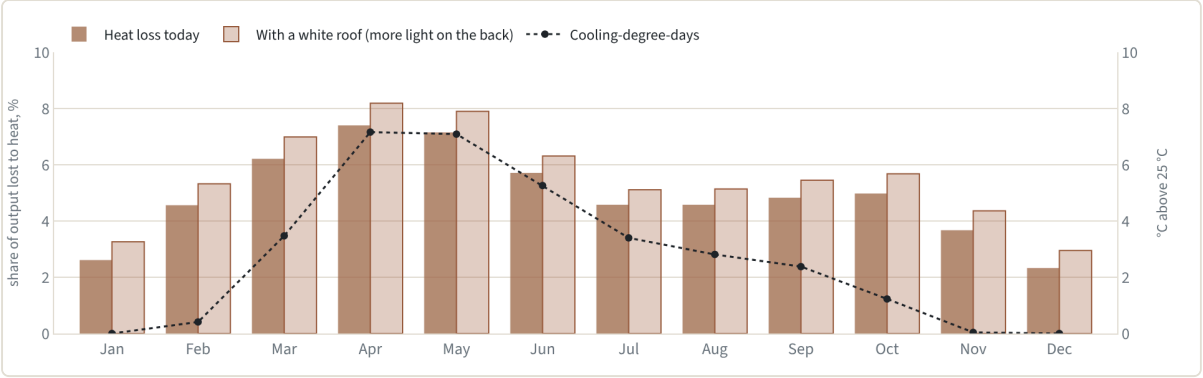
All nine bills together. Left: bill offset is the green and gold slices together; self-sufficiency is the green slice alone. Right: the same solar, split by where it went.

Two percentages are easy to confuse, and the left-hand chart shows both. Of all the electricity the house used, 26% came from the panels at the moment they were producing; that green slice is **self-sufficiency**. Another 20% came from the grid but was cancelled on the bill by units the panels had exported earlier; that is the gold slice. Together they make the **bill offset**: generation equal to 46% of household demand. The remaining 54% was bought and paid for. The right-hand chart shows the same solar from the other side: 57% was used at home and 43% went to the grid.

A cheap experiment: paint the roof white

These panels are bifacial: they collect light on the back as well as the front, from sunlight reflected off the surface below. Today that surface is dark, often damp concrete, which reflects roughly 12% of the light falling on it. White paint reflects about 70%. The module datasheet's rear-gain table runs from 15% to 30% depending on the ground [6], and on the current surface the rear side probably adds about 5%. Our estimate for a white surface is about 12% more light-driven output.

Would it also keep the panels cooler in summer? This was worth checking, because heat is what costs these panels most in exactly the months with the highest cooling-degree-days. In a typical year, heat takes 7.4% of output in April, when the cells run about 18 °C above the air, against 2.3% in December.



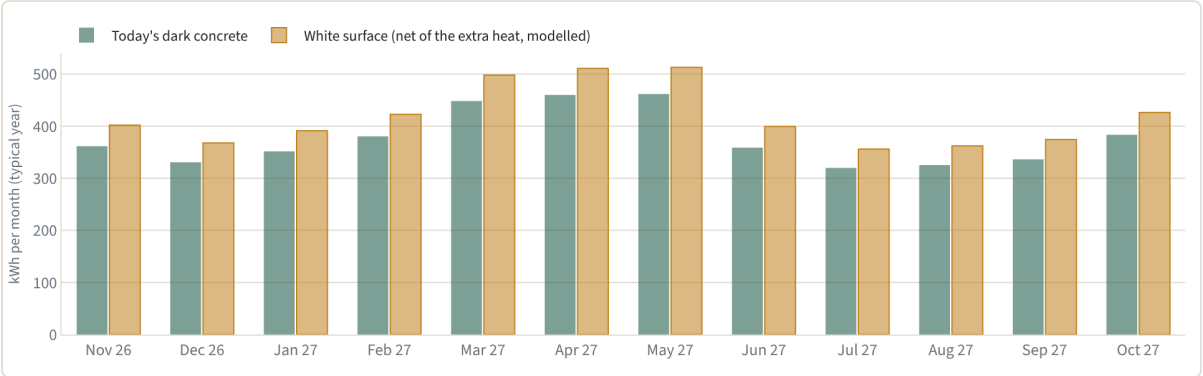
Share of output lost to panel heat in a typical year, today and with a white roof below, against cooling-degree-days. Hover for how far the cells run above the air.

Probably not, though the evidence here is thinner than for the light gain. Two effects pull in opposite directions.

- **The back of the panel gets hotter.** Reflected light that reaches the back is absorbed, and absorbed light that does not become electricity becomes heat. With bifaciality of 80% [6], a 12% output gain means about 15% more light landing on the module. In our model, which applies the NOCT relation to that extra light, the cells run about 2.6 °C warmer in April and the heat loss rises from 7.4% to 8.2%. This is a model result, not a measurement.
- **The surroundings get cooler.** A white roof stays cooler in the sun [8], so it radiates less heat at the panels and warms the air around them less. Over a cool-roof coating, monofacial panels, which cannot use the reflected light and so show only this effect, ran 1.3 °C cooler [9].

So at the summer peak the cells end up between 1.3 °C warmer (if the cooling measured on another roof applies here) and 2.6 °C warmer (if it does not). We found no published test that measures bifacial cell temperature over a dark roof and a white one side by side. What the field tests agree on is that the energy gain from a reflective roof comes mostly from extra light, not from lower temperatures [10][8], and the bifacial gain is much larger than the monofacial one [9]. For this array the paint boosts the light. It is not a coolant.

The heat costs between 16 and 36 kWh a year, depending on which bound holds. Either way that is small next to the 542 kWh of extra light.

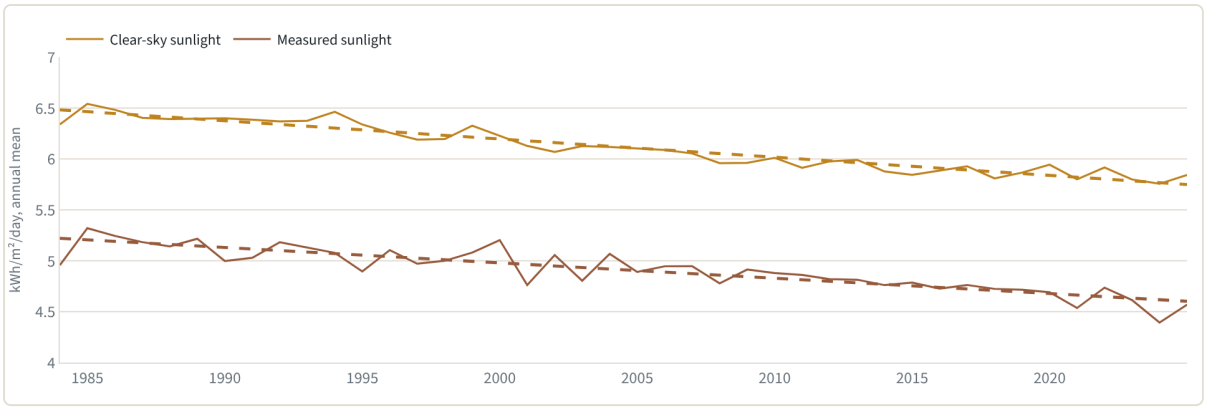


Typical-year generation with today's dark concrete and with a white surface beneath the panels, after subtracting the extra heat. Modelled from the module datasheet, not measured on this roof.

Net of that heat, a white surface adds 506 to 527 kWh in a typical year (we use the lower figure), roughly 11%, worth up to ₹3,200 a year at the top tariff slab of ₹6.34 per unit [7]. Winter units fall in cheaper slabs, so treat that as a ceiling. On ₹8,000 to ₹12,000 of paint and labour, the payback is roughly 2.5 to 3.7 years. Paint on the parts of the roof not covered by panels could also keep the rooms below a little cooler in summer [8], but we have no data to put a number on that for this house.

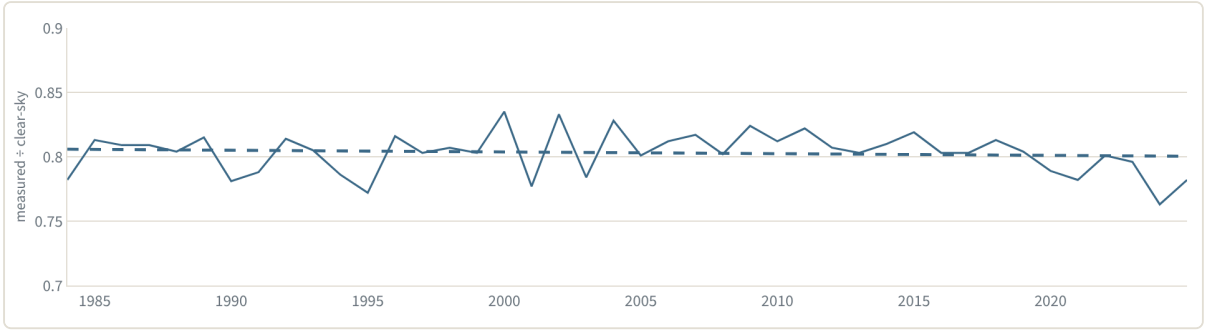
The sky is getting darker

The last hunch came from a longer view. If sunlight drives everything, is the sunlight itself changing? NASA's record for these coordinates goes back to 1984.

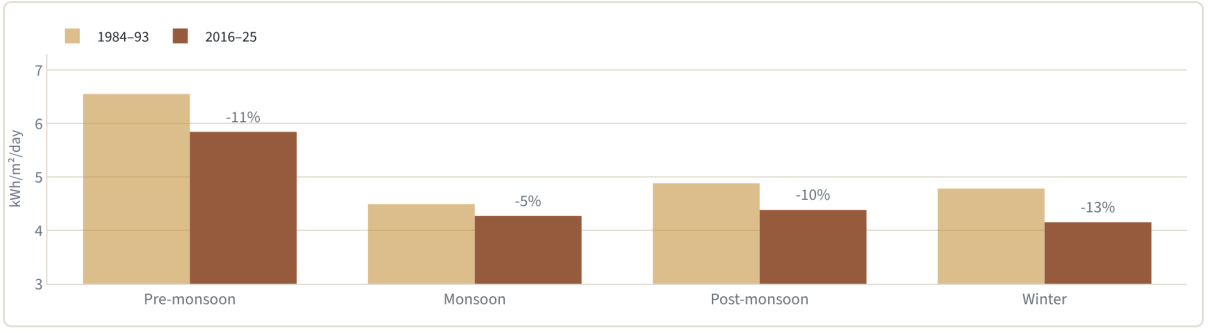


Annual mean sunlight at these coordinates, 1984–2025: what reached the ground (measured) and what would on a cloudless day (clear-sky). Dashed: trends.

Average sunlight fell from 5.14 kWh/m² per day in 1984–93 to 4.65 in 2016–25, a decline of −0.15 per decade ($p < 0.0001$). Ground stations have recorded the same dimming across India for decades [11][12]. The natural suspect is cloud. But at these coordinates the clear-sky series, the sunlight a cloudless day would deliver, fell even faster, at −0.18 per decade.



Clearness index, measured ÷ clear-sky. Flat means cloud cover has not changed.

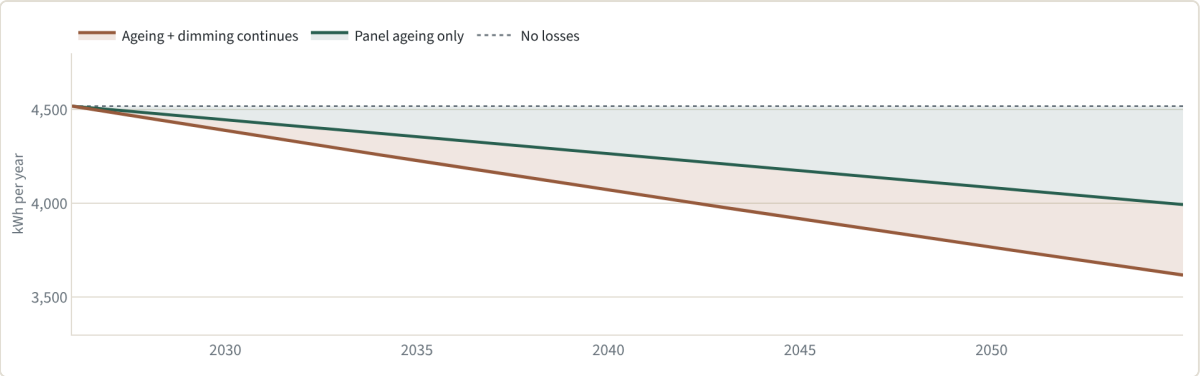


Seasonal sunlight in the first and the latest decade of the record.

The clearness index, measured sunlight divided by clear-sky sunlight, is flat (−0.001 per decade, $p = 0.53$). Over this grid cell, cloud cover has not changed; the air itself has become less transparent. That fits a 2024 study that found clouds dominate dimming across India as a whole, but named the region containing Bhubaneswar as the country's most aerosol-affected [13]. The dry seasons lost the most here, and those are the months in which a rooftop array earns most of its keep.

The same NASA record shows daily maximum temperatures falling, while India's own station network shows them rising [14]. We treat that as an artefact of the dataset and build nothing on it.

Over the panels' life



Annual output over the panels' 30-year power warranty. Ageing follows the datasheet warranty line; dimming extrapolates the 42-year trend, a downside case rather than a forecast.

Two slow losses compound. The panels age: the datasheet warrants at least 99% of rated power after the first year and 87.4% after 30 years, about 0.4% a year [6]. And if the dimming trend continues at 0.32% a year, sunlight falls further over the same period. By 2055 the array would make about 3,617 kWh, 20% below year one, about 8 percentage points worse than ageing alone. Aerosol pollution is a policy choice rather than a law of nature. Europe has brightened since the 1980s as air pollution was cut [15][16], and eastern China has brightened since its 2013 Clean Air Action [17], after decades in which pollution had measurably cut solar output [18]. The lower line is a downside case, not a forecast.

How we worked

- **Data.** Nine TPCODL bills, 8 February 2026 to 1 November 2026, reconciled against cumulative meter readings. NASA POWER daily weather for the billed days, ten years of daily history (2016 to 2025), and the monthly and annual record back to 1984 [1].
- **Model one, for forecasting.** Generation per day = a × sunlight + b, fitted by least squares on the nine months.
- **Model two, for diagnosis and the typical-year forecast.** Sunlight on the panels × heat derate × kWp × η, with η the only fitted number.
- **Validation.** Two genuine out-of-sample months (September, then October against a published forecast) and leave-one-out across all nine. A pre-registered winter test of three explanations for the residuals, scored against a frozen model.
- **Assumptions.** Panels fixed at 12° facing south. Ground reflectivity 0.12. Cell temperature from a NOCT of 43 °C. Tariff slabs of ₹2.90, 4.70, 5.70 and 6.10 per unit plus 4% duty. The house empty for 7.5 days in May and guests for about a week in October (demand only). White-surface gain of 12% from the datasheet's rear-gain table.
- **Limits.** Nine monthly points. An efficiency fitted on the same months it judges. Satellite estimates for a grid cell rather than a sensor on the roof. No inverter-level data.

Every number and chart in this piece is regenerated from the project's data with a single command.

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